In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
df = pd.DataFrame({

    "Ride_ID": [
        "R001", "R002", "R003", "R001", "R005",
        "R006", "R007", "R008", "R009", "R010",
        "R011", "R012", "R013", "R014", "R015",
        "R016", "R017", "R018", "R019", "R020"
    ],

    "Bike_Type": [
        "Electric", "Standard", "Mountain", "Electric",
        "Hybrid", "standard", "Electric", "Mountain",
        "Hybrid", "Electric", "Standard", "Mountain",
        "Electric", "Hybrid", "Mountain", "Electric",
        "Standard", "Hybrid", "Mountain", "Electric"
    ],

    "Ride_Date": [
        "2026-01-05", "2026-01-08", "2026-01-12",
        "2026-01-05", "2026-01-15", "2026-01-18",
        "2026-01-20", "2026-01-22", "2026-01-25",
        "2026-01-28", "2026-02-01", "2026-02-04",
        "2026-02-07", "2026-02-10", "2026-02-13",
        "2026-02-16", "2026-02-19", "2026-02-22",
        "2026-02-25", "invalid-date"
    ],

    "Area": [
        "Hyderabad", "Chennai", "Bangalore", "Hyderabad",
        "Mumbai", "Pune", "Delhi", "Chennai", "Mumbai",
        "Bangalore", "Hyderabad", "Pune", "Delhi",
        "Mumbai", "Chennai", "Bangalore", "Pune",
        "Delhi", "Mumbai", "Hyderabad"
    ],

    "Distance_Km": [
        5.2, 3.8, 12.5, 5.2, 8.4,
        2.5, 15.6, 7.8, 4.2, 9.5,
        6.3, 11.2, 0.8, 14.5, 5.9,
        999999, 4.7, 8.9, 3.2, "unknown"
    ],

    "Duration_Min": [
        25, 18, 45, 25, 35,
        15, 60, np.nan, 22, 40,
        30, 55, 10, 65, 28,
        75, -10, 38, 20, 32
    ],
```

```
    "Fare": [
        120, 90, 250, 120, 180,
        75, 300, 150, np.nan, 220,
        140, 275, 50, 320, 160,
        1500, 110, 200, 95, 175
    ],

    "Status": [
        "Completed", "Completed", "Active", "Completed",
        "Completed", "active", "Completed", "Pending",
        "Completed", "Active", "Completed", "Pending",
        "Completed", "Cancelled", "Completed",
        "Active", "Completed", "Pending", " Completed ",
        "Unknown"
    ]
})

df
```

Out[2]:

	Ride_ID	Bike_Type	Ride_Date	Area	Distance_Km	Duration_Min	Fare	Sta
0	R001	Electric	2026-01-05	Hyderabad	5.2	25.0	120.0	Comple
1	R002	Standard	2026-01-08	Chennai	3.8	18.0	90.0	Comple
2	R003	Mountain	2026-01-12	Bangalore	12.5	45.0	250.0	Ac
3	R001	Electric	2026-01-05	Hyderabad	5.2	25.0	120.0	Comple
4	R005	Hybrid	2026-01-15	Mumbai	8.4	35.0	180.0	Comple
5	R006	standard	2026-01-18	Pune	2.5	15.0	75.0	ac
6	R007	Electric	2026-01-20	Delhi	15.6	60.0	300.0	Comple
7	R008	Mountain	2026-01-22	Chennai	7.8	NaN	150.0	Penc
8	R009	Hybrid	2026-01-25	Mumbai	4.2	22.0	NaN	Comple
9	R010	Electric	2026-01-28	Bangalore	9.5	40.0	220.0	Ac
10	R011	Standard	2026-02-01	Hyderabad	6.3	30.0	140.0	Comple
11	R012	Mountain	2026-02-04	Pune	11.2	55.0	275.0	Penc
12	R013	Electric	2026-02-07	Delhi	0.8	10.0	50.0	Comple
13	R014	Hybrid	2026-02-10	Mumbai	14.5	65.0	320.0	Cance
14	R015	Mountain	2026-02-13	Chennai	5.9	28.0	160.0	Comple
15	R016	Electric	2026-02-16	Bangalore	999999	75.0	1500.0	Ac
16	R017	Standard	2026-02-19	Pune	4.7	-10.0	110.0	Comple
17	R018	Hybrid	2026-02-22	Delhi	8.9	38.0	200.0	Penc
18	R019	Mountain	2026-02-25	Mumbai	3.2	20.0	95.0	Comple

	Ride_ID	Bike_Type	Ride_Date	Area	Distance_Km	Duration_Min	Fare	Sta
19	R020	Electric	invalid-date	Hyderabad	unknown	32.0	175.0	Unkn

In [3]:
```
df.head()
```

Out[3]:

	Ride_ID	Bike_Type	Ride_Date	Area	Distance_Km	Duration_Min	Fare	Statu
0	R001	Electric	2026-01-05	Hyderabad	5.2	25.0	120.0	Complete
1	R002	Standard	2026-01-08	Chennai	3.8	18.0	90.0	Complete
2	R003	Mountain	2026-01-12	Bangalore	12.5	45.0	250.0	Activ
3	R001	Electric	2026-01-05	Hyderabad	5.2	25.0	120.0	Complete
4	R005	Hybrid	2026-01-15	Mumbai	8.4	35.0	180.0	Complete

In [4]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 8 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   Ride_ID       20 non-null     str    
 1   Bike_Type     20 non-null     str    
 2   Ride_Date     20 non-null     str    
 3   Area          20 non-null     str    
 4   Distance_Km   20 non-null     object 
 5   Duration_Min  19 non-null     float64
 6   Fare          19 non-null     float64
 7   Status        20 non-null     str    
dtypes: float64(2), object(1), str(5)
memory usage: 1.4+ KB
```

In [5]:
```
df.shape
```

Out[5]:
```
(20, 8)
```

In [6]:
```
df.describe
```

Out[6]:
```
<bound method DataFrame.describe of    Ride_ID Bike_Type     Ride_Date       Area
Distance_Km  Duration_Min  \
0     R001  Electric    2026-01-05  Hyderabad         5.2          25.0
1     R002  Standard    2026-01-08    Chennai         3.8          18.0
2     R003  Mountain    2026-01-12  Bangalore        12.5          45.0
3     R001  Electric    2026-01-05  Hyderabad         5.2          25.0
4     R005    Hybrid    2026-01-15     Mumbai         8.4          35.0
5     R006  standard    2026-01-18       Pune         2.5          15.0
6     R007  Electric    2026-01-20      Delhi        15.6          60.0
7     R008  Mountain    2026-01-22    Chennai         7.8           NaN
8     R009    Hybrid    2026-01-25     Mumbai         4.2          22.0
9     R010  Electric    2026-01-28  Bangalore         9.5          40.0
10    R011  Standard    2026-02-01  Hyderabad         6.3          30.0
11    R012  Mountain    2026-02-04       Pune        11.2          55.0
12    R013  Electric    2026-02-07      Delhi         0.8          10.0
13    R014    Hybrid    2026-02-10     Mumbai        14.5          65.0
14    R015  Mountain    2026-02-13    Chennai         5.9          28.0
15    R016  Electric    2026-02-16  Bangalore      999999          75.0
16    R017  Standard    2026-02-19       Pune         4.7         -10.0
17    R018    Hybrid    2026-02-22      Delhi         8.9          38.0
18    R019  Mountain    2026-02-25     Mumbai         3.2          20.0
19    R020  Electric  invalid-date  Hyderabad     unknown          32.0

      Fare       Status
0    120.0    Completed
1     90.0    Completed
2    250.0       Active
3    120.0    Completed
4    180.0    Completed
5     75.0       active
6    300.0    Completed
7    150.0      Pending
8      NaN    Completed
9    220.0       Active
10   140.0    Completed
11   275.0      Pending
12    50.0    Completed
13   320.0    Cancelled
14   160.0    Completed
15  1500.0       Active
16   110.0    Completed
17   200.0      Pending
18    95.0   Completed
19   175.0      Unknown  >
```

In [7]:
```
df.isnull().sum()
```

Out[7]:
```
Ride_ID         0
Bike_Type       0
Ride_Date       0
Area            0
Distance_Km     0
Duration_Min    1
Fare            1
Status          0
dtype: int64
```

In [8]:
```
df.isna().sum()
```

Out[8]:
```
Ride_ID         0
Bike_Type       0
Ride_Date       0
Area            0
Distance_Km     0
Duration_Min    1
Fare            1
Status          0
dtype: int64
```

In [9]:
```
## Fill missing duration values with median

df["Duration_Min"] = df["Duration_Min"].fillna(
    df["Duration_Min"].median()
)

df["Duration_Min"]
```

Out[9]:
```
0     25.0
1     18.0
2     45.0
3     25.0
4     35.0
5     15.0
6     60.0
7     30.0
8     22.0
9     40.0
10    30.0
11    55.0
12    10.0
13    65.0
14    28.0
15    75.0
16   -10.0
17    38.0
18    20.0
19    32.0
Name: Duration_Min, dtype: float64
```

In [10]:
```
## Fill missing fare values with median

df["Fare"] = df["Fare"].fillna(
    df["Fare"].median()
```

```
)

df["Fare"]
```

Out[10]:
```
0      120.0
1       90.0
2      250.0
3      120.0
4      180.0
5       75.0
6      300.0
7      150.0
8      160.0
9      220.0
10     140.0
11     275.0
12      50.0
13     320.0
14     160.0
15    1500.0
16     110.0
17     200.0
18      95.0
19     175.0
Name: Fare, dtype: float64
```

In [11]:
```
df = df.drop_duplicates()

print(df.shape)
```

```
(19, 8)
```

In [12]:
```
df["Bike_Type"] = df["Bike_Type"].str.strip().str.title()

df["Status"] = df["Status"].str.strip().str.title()

df
```

Out[12]:

	Ride_ID	Bike_Type	Ride_Date	Area	Distance_Km	Duration_Min	Fare	Sta
0	R001	Electric	2026-01-05	Hyderabad	5.2	25.0	120.0	Comple
1	R002	Standard	2026-01-08	Chennai	3.8	18.0	90.0	Comple
2	R003	Mountain	2026-01-12	Bangalore	12.5	45.0	250.0	Ac
4	R005	Hybrid	2026-01-15	Mumbai	8.4	35.0	180.0	Comple
5	R006	Standard	2026-01-18	Pune	2.5	15.0	75.0	Ac
6	R007	Electric	2026-01-20	Delhi	15.6	60.0	300.0	Comple
7	R008	Mountain	2026-01-22	Chennai	7.8	30.0	150.0	Penc
8	R009	Hybrid	2026-01-25	Mumbai	4.2	22.0	160.0	Comple
9	R010	Electric	2026-01-28	Bangalore	9.5	40.0	220.0	Ac
10	R011	Standard	2026-02-01	Hyderabad	6.3	30.0	140.0	Comple
11	R012	Mountain	2026-02-04	Pune	11.2	55.0	275.0	Penc
12	R013	Electric	2026-02-07	Delhi	0.8	10.0	50.0	Comple
13	R014	Hybrid	2026-02-10	Mumbai	14.5	65.0	320.0	Cance
14	R015	Mountain	2026-02-13	Chennai	5.9	28.0	160.0	Comple
15	R016	Electric	2026-02-16	Bangalore	999999	75.0	1500.0	Ac
16	R017	Standard	2026-02-19	Pune	4.7	-10.0	110.0	Comple
17	R018	Hybrid	2026-02-22	Delhi	8.9	38.0	200.0	Penc
18	R019	Mountain	2026-02-25	Mumbai	3.2	20.0	95.0	Comple
19	R020	Electric	invalid-date	Hyderabad	unknown	32.0	175.0	Unkno

In [13]:
```
df[["Bike_Type", "Area", "Status"]].head(10)
```

Out[13]:

	Bike_Type	Area	Status
0	Electric	Hyderabad	Completed
1	Standard	Chennai	Completed
2	Mountain	Bangalore	Active
4	Hybrid	Mumbai	Completed
5	Standard	Pune	Active
6	Electric	Delhi	Completed
7	Mountain	Chennai	Pending
8	Hybrid	Mumbai	Completed
9	Electric	Bangalore	Active
10	Standard	Hyderabad	Completed

In [14]:
```
## Convert Distance_Km to numeric

df["Distance_Km"] = pd.to_numeric(
    df["Distance_Km"], errors="coerce"
)

df["Distance_Km"]
```

Out[14]:
```
0          5.2
1          3.8
2         12.5
4          8.4
5          2.5
6         15.6
7          7.8
8          4.2
9          9.5
10         6.3
11        11.2
12         0.8
13        14.5
14         5.9
15    999999.0
16         4.7
17         8.9
18         3.2
19         NaN
Name: Distance_Km, dtype: float64
```

In [15]:
```
## Convert Ride_Date to datetime

df["Ride_Date"] = pd.to_datetime(
```

```
    df["Ride_Date"], errors="coerce"
)

df.info()
```

```
<class 'pandas.DataFrame'>
Index: 19 entries, 0 to 19
Data columns (total 8 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Ride_ID       19 non-null     str
 1   Bike_Type     19 non-null     str
 2   Ride_Date     18 non-null     datetime64[us]
 3   Area          19 non-null     str
 4   Distance_Km   18 non-null     float64
 5   Duration_Min  19 non-null     float64
 6   Fare          19 non-null     float64
 7   Status        19 non-null     str
dtypes: datetime64[us](1), float64(3), str(4)
memory usage: 1.3 KB
```

In [16]:
```
Q1 = df["Distance_Km"].quantile(0.25)
Q3 = df["Distance_Km"].quantile(0.75)

IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower bound:", lower_bound)
print("Upper bound:", upper_bound)
```

```
Q1: 4.325
Q3: 10.774999999999999
IQR: 6.449999999999998
Lower bound: -5.349999999999997
Upper bound: 20.449999999999996
```

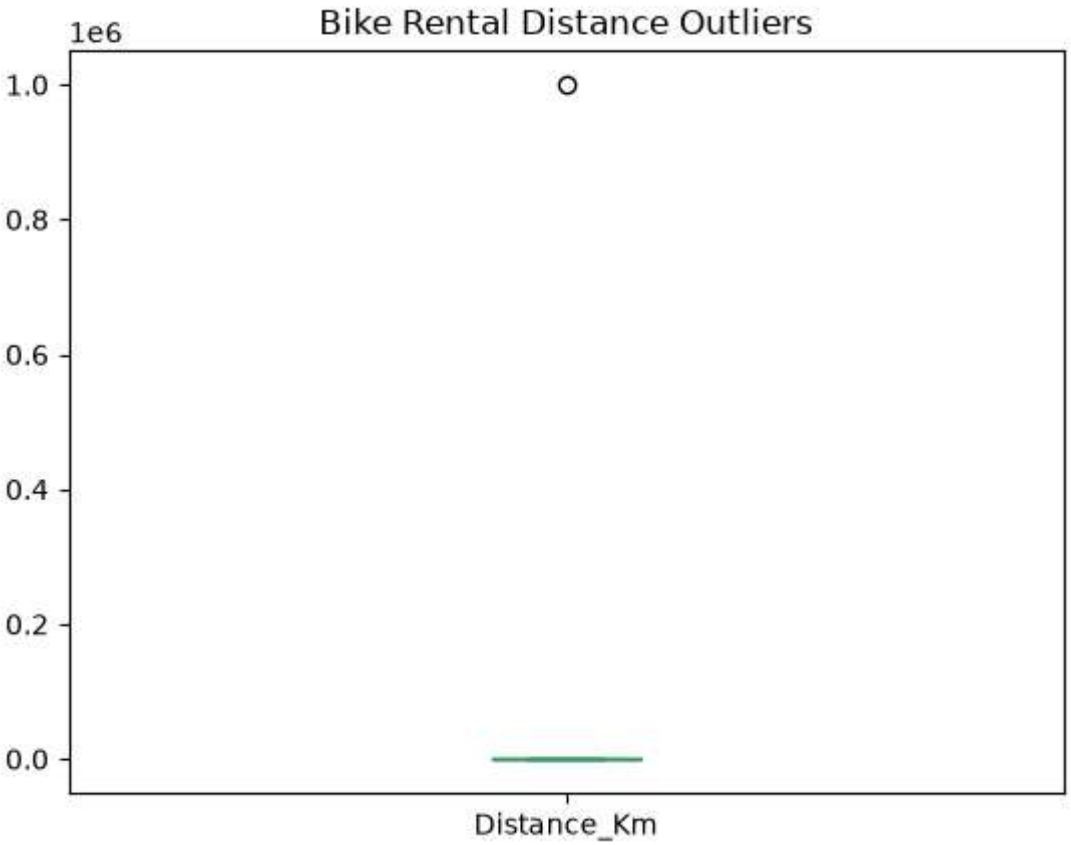
In [17]:
```
outliers = df[
    (df["Distance_Km"] < lower_bound) |
    (df["Distance_Km"] > upper_bound)
]

display(outliers)
```

	Ride_ID	Bike_Type	Ride_Date	Area	Distance_Km	Duration_Min	Fare	Status
15	R016	Electric	2026-02-16	Bangalore	999999.0	75.0	1500.0	Active

In [18]:
```
df["Distance_Km"].plot(kind="box", title="Bike Rental Distance Outliers")

plt.show()
```



In [19]:
```
df[df["Duration_Min"] < 0]
```

Out[19]:

	Ride_ID	Bike_Type	Ride_Date	Area	Distance_Km	Duration_Min	Fare	Status
16	R017	Standard	2026-02-19	Pune	4.7	-10.0	110.0	Completed

In [20]:
```
median_duration = df.loc[
    df["Duration_Min"] >= 0,
    "Duration_Min"
].median()

df.loc[
    df["Duration_Min"] < 0,
    "Duration_Min"
] = median_duration

df["Duration_Min"]
```

Out[20]:
```
0     25.0
1     18.0
2     45.0
4     35.0
5     15.0
6     60.0
7     30.0
8     22.0
9     40.0
10    30.0
11    55.0
12    10.0
13    65.0
14    28.0
15    75.0
16    31.0
17    38.0
18    20.0
19    32.0
Name: Duration_Min, dtype: float64
```

In [21]:
```
df["Ride_Year"] = df["Ride_Date"].dt.year

df["Ride_Year"]
```

Out[21]:
```
0     2026.0
1     2026.0
2     2026.0
4     2026.0
5     2026.0
6     2026.0
7     2026.0
8     2026.0
9     2026.0
10    2026.0
11    2026.0
12    2026.0
13    2026.0
14    2026.0
15    2026.0
16    2026.0
17    2026.0
18    2026.0
19       NaN
Name: Ride_Year, dtype: float64
```

In [22]:
```
df["Ride_Month"] = df["Ride_Date"].dt.month

df[["Ride_Year", "Ride_Month"]]
```

Out[22]:

	Ride_Year	Ride_Month
0	2026.0	1.0
1	2026.0	1.0
2	2026.0	1.0
4	2026.0	1.0
5	2026.0	1.0
6	2026.0	1.0
7	2026.0	1.0
8	2026.0	1.0
9	2026.0	1.0
10	2026.0	2.0
11	2026.0	2.0
12	2026.0	2.0
13	2026.0	2.0
14	2026.0	2.0
15	2026.0	2.0
16	2026.0	2.0
17	2026.0	2.0
18	2026.0	2.0
19	NaN	NaN

In [23]:

```
## Classify bike ride distance

def classify_distance(distance):

    if pd.isna(distance):
        return "Unknown"

    elif distance < 5:
        return "Short Distance"

    elif distance < 10:
        return "Medium Distance"

    else:
        return "Long Distance"


df["Distance_Category"] = df["Distance_Km"].apply(
    classify_distance
```

```
)

df[["Distance_Km", "Distance_Category"]]
```

Out[23]:

	Distance_Km	Distance_Category
0	5.2	Medium Distance
1	3.8	Short Distance
2	12.5	Long Distance
4	8.4	Medium Distance
5	2.5	Short Distance
6	15.6	Long Distance
7	7.8	Medium Distance
8	4.2	Short Distance
9	9.5	Medium Distance
10	6.3	Medium Distance
11	11.2	Long Distance
12	0.8	Short Distance
13	14.5	Long Distance
14	5.9	Medium Distance
15	999999.0	Long Distance
16	4.7	Short Distance
17	8.9	Medium Distance
18	3.2	Short Distance
19	NaN	Unknown

In [24]:
```
df[~df["Status"].isin(
    ["Active", "Completed", "Pending", "Cancelled"]
)]
```

Out[24]:

	Ride_ID	Bike_Type	Ride_Date	Area	Distance_Km	Duration_Min	Fare	Statu
19	R020	Electric	NaT	Hyderabad	NaN	32.0	175.0	Unknowı

In [25]:
```
df = df.dropna(
    subset=["Ride_Date", "Distance_Km"]
)

df
```

Out[25]:

	Ride_ID	Bike_Type	Ride_Date	Area	Distance_Km	Duration_Min	Fare	Sta
0	R001	Electric	2026-01-05	Hyderabad	5.2	25.0	120.0	Comple
1	R002	Standard	2026-01-08	Chennai	3.8	18.0	90.0	Comple
2	R003	Mountain	2026-01-12	Bangalore	12.5	45.0	250.0	Ac
4	R005	Hybrid	2026-01-15	Mumbai	8.4	35.0	180.0	Comple
5	R006	Standard	2026-01-18	Pune	2.5	15.0	75.0	Ac
6	R007	Electric	2026-01-20	Delhi	15.6	60.0	300.0	Comple
7	R008	Mountain	2026-01-22	Chennai	7.8	30.0	150.0	Penc
8	R009	Hybrid	2026-01-25	Mumbai	4.2	22.0	160.0	Comple
9	R010	Electric	2026-01-28	Bangalore	9.5	40.0	220.0	Ac
10	R011	Standard	2026-02-01	Hyderabad	6.3	30.0	140.0	Comple
11	R012	Mountain	2026-02-04	Pune	11.2	55.0	275.0	Penc
12	R013	Electric	2026-02-07	Delhi	0.8	10.0	50.0	Comple
13	R014	Hybrid	2026-02-10	Mumbai	14.5	65.0	320.0	Cance
14	R015	Mountain	2026-02-13	Chennai	5.9	28.0	160.0	Comple
15	R016	Electric	2026-02-16	Bangalore	999999.0	75.0	1500.0	Ac
16	R017	Standard	2026-02-19	Pune	4.7	31.0	110.0	Comple
17	R018	Hybrid	2026-02-22	Delhi	8.9	38.0	200.0	Penc
18	R019	Mountain	2026-02-25	Mumbai	3.2	20.0	95.0	Comple

In [26]:
```
df.isnull().sum()
```

Out[26]:
```
Ride_ID             0
Bike_Type           0
Ride_Date           0
Area                0
Distance_Km         0
Duration_Min        0
Fare                0
Status              0
Ride_Year           0
Ride_Month          0
Distance_Category   0
dtype: int64
```

In [27]:
```
df.groupby("Area")["Fare"].sum()
```

Out[27]:
```
Area
Bangalore    1970.0
Chennai       400.0
Delhi         550.0
Hyderabad     260.0
Mumbai        755.0
Pune          460.0
Name: Fare, dtype: float64
```

In [28]:
```
df.groupby("Bike_Type")["Distance_Km"].mean()
```

Out[28]:
```
Bike_Type
Electric    200006.020
Hybrid           9.000
Mountain         8.120
Standard         4.325
Name: Distance_Km, dtype: float64
```

In [29]:
```
df.groupby("Status")["Ride_ID"].count()
```

Out[29]:
```
Status
Active        4
Cancelled     1
Completed    10
Pending       3
Name: Ride_ID, dtype: int64
```

In [30]:
```
df.groupby("Area")["Fare"].agg(
    ["sum", "mean", "min", "max", "count"]
)
```

Out[30]:

Area	sum	mean	min	max	count
Bangalore	1970.0	656.666667	220.0	1500.0	3
Chennai	400.0	133.333333	90.0	160.0	3
Delhi	550.0	183.333333	50.0	300.0	3
Hyderabad	260.0	130.000000	120.0	140.0	2
Mumbai	755.0	188.750000	95.0	320.0	4
Pune	460.0	153.333333	75.0	275.0	3

In [31]:
```
df.groupby(
    ["Area", "Bike_Type"]
)["Fare"].sum().reset_index()
```

Out[31]:

	Area	Bike_Type	Fare
0	Bangalore	Electric	1720.0
1	Bangalore	Mountain	250.0
2	Chennai	Mountain	310.0
3	Chennai	Standard	90.0
4	Delhi	Electric	350.0
5	Delhi	Hybrid	200.0
6	Hyderabad	Electric	120.0
7	Hyderabad	Standard	140.0
8	Mumbai	Hybrid	660.0
9	Mumbai	Mountain	95.0
10	Pune	Mountain	275.0
11	Pune	Standard	185.0

In [32]:
```
df.shape
```

Out[32]:
```
(18, 11)
```

In [33]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
Index: 18 entries, 0 to 18
Data columns (total 11 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Ride_ID            18 non-null     str
 1   Bike_Type          18 non-null     str
 2   Ride_Date          18 non-null     datetime64[us]
 3   Area               18 non-null     str
 4   Distance_Km        18 non-null     float64
 5   Duration_Min       18 non-null     float64
 6   Fare               18 non-null     float64
 7   Status             18 non-null     str
 8   Ride_Year          18 non-null     float64
 9   Ride_Month         18 non-null     float64
 10  Distance_Category  18 non-null     str
dtypes: datetime64[us](1), float64(5), str(5)
memory usage: 1.7 KB
```

In [34]:
```
## Save cleaned dataset

df.to_csv("bike_rental_cleaned.csv", index=False)

print("Cleaned dataset saved successfully!")
```

```
Cleaned dataset saved successfully!
```

In [35]:
```
df.head(10)
```

Out[35]:

	Ride_ID	Bike_Type	Ride_Date	Area	Distance_Km	Duration_Min	Fare	Stat
0	R001	Electric	2026-01-05	Hyderabad	5.2	25.0	120.0	Complet
1	R002	Standard	2026-01-08	Chennai	3.8	18.0	90.0	Complet
2	R003	Mountain	2026-01-12	Bangalore	12.5	45.0	250.0	Acti
4	R005	Hybrid	2026-01-15	Mumbai	8.4	35.0	180.0	Complet
5	R006	Standard	2026-01-18	Pune	2.5	15.0	75.0	Acti
6	R007	Electric	2026-01-20	Delhi	15.6	60.0	300.0	Complet
7	R008	Mountain	2026-01-22	Chennai	7.8	30.0	150.0	Pendi
8	R009	Hybrid	2026-01-25	Mumbai	4.2	22.0	160.0	Complet
9	R010	Electric	2026-01-28	Bangalore	9.5	40.0	220.0	Acti
10	R011	Standard	2026-02-01	Hyderabad	6.3	30.0	140.0	Complet

In [ ]: